Classifying books; its weight in grams and price in peso. Thin paperbacks and pocketbooks—lightweight and cheap; medium weight and mid-range price for standard novels and textbooks; and thick hardcovers, anthologies, and encyclopaedias: heavy and expensive. To identify these natural groupings objectively, K-Means clustering was applied to a dataset of 30 books using two features: weight in grams and price in Philippine Peso.

Before running the final clustering, the optimal number of clusters k was determined using the Elbow Method. K-Means was run for k = 1 through k = 10, and the Within-Cluster Sum of Squares (WCSS) was recorded for each run. The following python snippet shows this process:

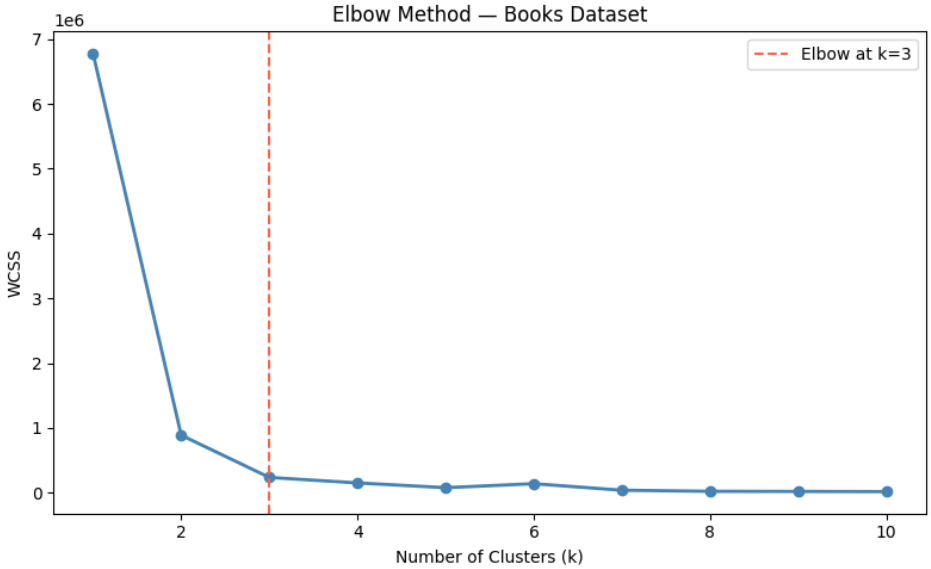
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

data = [
    [120,24],[150,30],[100,20],[180,36],[130,26],[160,32],
    [450,108],[500,120],[420,101],[550,132],[480,115],[520,125],
    [460,110],[530,127],[400,96],[1200,264],[1350,270],[1100,242],
    [1400,280],[1250,266],[1300,273],[1280,269],[1380,276],[1150,253],
    [1420,284],[220,44],[250,50],[200,40],[680,163],[720,173]
]

wcss = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, init='k-means++', random_state=42)
    km.fit(data)
    wcss.append(km.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS')
plt.show()
```

The elbow was found at **k = 3**, confirming that three clusters best represent the structure of the data. Initial centroids were then selected using the K-means++ strategy to ensure well-spread starting positions before proceeding to the manual iteration tables.



K-means Algorithm on a Bookstore Book Classification		
#	Weight (g)	Price (PHP)
1	120	24
2	150	30
3	100	20
4	180	36
5	130	26
6	160	32
7	450	108
8	500	120
9	420	101
10	550	132
11	480	115
12	520	125
13	460	110
14	530	127
15	400	96
16	1200	264
17	1350	270
18	1100	242
19	1400	280
20	1250	266
21	1300	273
22	1280	269
23	1380	276
24	1150	253
25	1420	284
26	220	44
27	250	50
28	200	40

29	680	163
30	720	173

The initial centroids were selected using the K-means++ initialization strategy to ensure a well-spread starting placement across the dataset. The first centroid, C1 = (400, 96)**,** was chosen as the starting point, representing a mid-range book in terms of both weight and price. Following the K-means++ rule, the next centroid was selected by identifying the point farthest from C1 — this yielded C2 = (1420, 284)**;** heavy and expensive end of the dataset. Finally, C3 = (100, 20) was selected as the point farthest from both C1 and C2;the lightest and cheapest books.

Iteration: 1						
#	Weight (x)	Price (y)	Distance to C1: (400, 96)	Distance to C2: (1420, 284)	Distance to C3: (100, 20)	Classification
1	120	24	289.08	1325.75	20.40	C3
2	150	30	258.57	1295.15	50.99	C3
3	100	20	309.48	1346.14	0	C3
4	180	36	228.04	1264.56	81.58	C3
5	130	26	278.93	1315.55	30.59	C3
6	160	32	248.39	1284.95	61.19	C3
7	450	108	51.42	985.84	360.89	C1
8	500	120	102.84	934.50	412.31	C1
9	420	101	20.62	1016.61	330.09	C1
10	550	132	154.26	883.18	463.73	C1
11	480	115	82.23	955.07	391.70	C1
12	520	125	123.45	913.94	432.93	C1
13	460	110	61.61	975.64	371.08	C1
14	530	127	133.65	903.74	443.11	C1
15	400	96	0.00	1037.18	309.48	C1
16	1200	264	817.45	220.91	1126.74	C2
17	1350	270	965.80	71.39	1274.75	C2
18	1100	242	715.06	322.74	1024.35	C2
19	1400	280	1016.79	20.40	1325.75	C2
20	1250	266	866.83	170.95	1176.02	C2
21	1300	273	917.24	120.50	1226.38	C2
22	1280	269	896.84	140.80	1205.99	C2
23	1380	276	996.39	40.79	1305.35	C2
24	1150	253	766.26	271.77	1075.54	C2
25	1420	284	1037.18	0.00	1346.14	C2
26	220	44	187.36	1223.76	122.38	C3
27	250	50	156.89	1193.17	152.97	C3
28	200	40	207.69	1244.16	101.98	C3
29	680	163	287.90	749.83	597.37	C1
30	720	173	329.13	708.75	683.60	C1

In Iteration 1, the Euclidean distance from each of the 30 data points to all three centroids was computed using the formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$, where x is the weight in grams and y

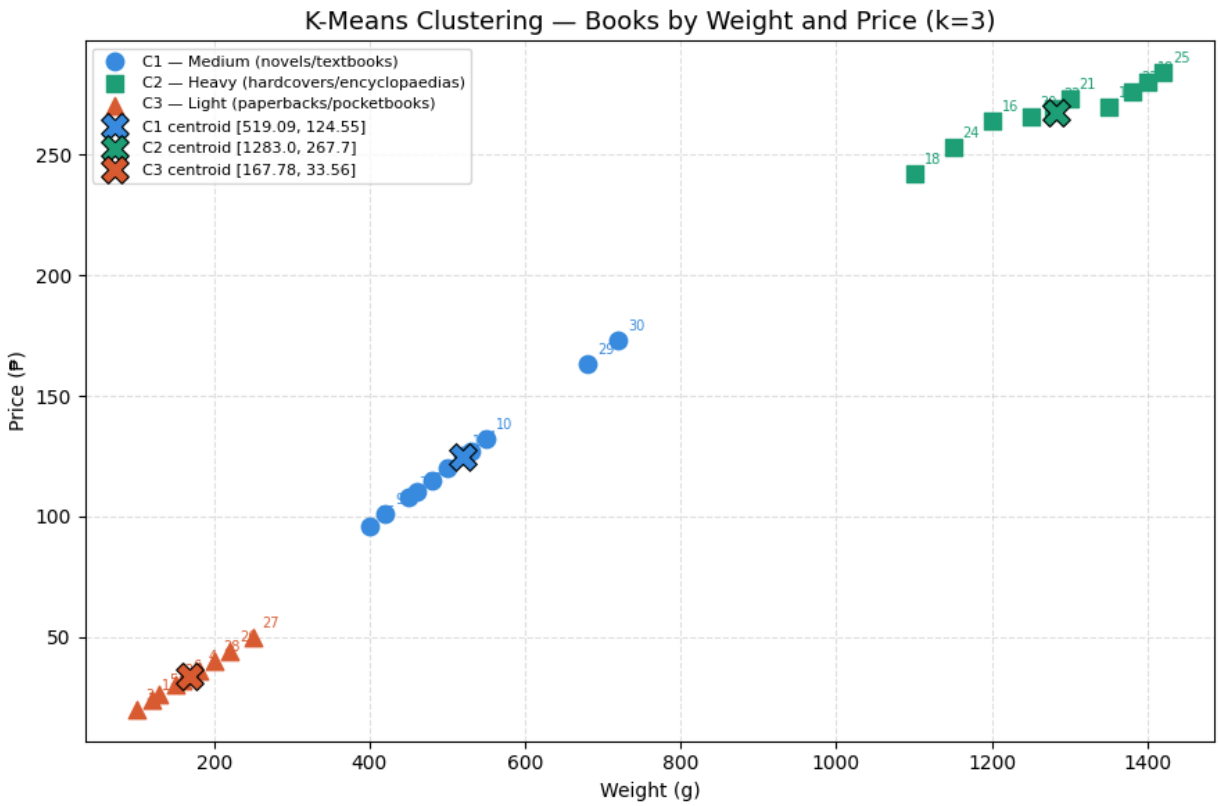
is the price in Philippine Peso. Each data point was then assigned to the nearest centroid—the one yielding the smallest distance, as shown in the Classification column.

Following the assignments—classifications, in Iteration 1, new centroids were computed by taking the mean weight and mean price of all points belonging to each cluster. The updated centroids are C1 **=** (519.10, 124.55)**,** C2 = (1283, 267.70), and C3 = (167.78, 33.56)**.**

Iteration: 2							
#	Weight (x)	Price (y)	Distance to C1: (519.10, 124.55)	Distance to C2: (1283, 267.70)	Distance to C3: (167.78, 33.56)	Classification: Iteration 1	Classification: Iteration 2
1	120	24	411.57	1188.26	48.73	C3	C3
2	150	30	381.02	1157.67	18.13	C3	C3
3	100	20	431.94	1208.65	69.12	C3	C3
4	180	36	350.47	1127.07	12.46	C3	C3
5	130	26	401.39	1178.06	38.53	C3	C3
6	160	32	370.83	1147.47	1.93	C3	C3
7	450	108	71.05	848.17	291.87	C1	C1
8	500	120	19.63	796.81	343.28	C1	C1
9	420	101	101.86	878.95	261.08	C1	C1
10	550	132	31.79	745.46	394.69	C1	C1
11	480	115	40.25	817.39	322.67	C1	C1
12	520	125	1.01	766.23	374.08	C1	C1
13	460	110	60.86	837.97	302.05	C1	C1
14	530	127	11.17	766.03	374.08	C1	C1
15	400	96	122.47	899.54	240.47	C1	C1
16	1200	264	695.03	83.08	1057.63	C2	C2
17	1350	270	843.53	67.04	1205.63	C2	C2
18	1100	242	592.65	184.80	955.24	C2	C2
19	1400	280	894.51	117.64	1256.62	C2	C2
20	1250	266	744.46	33.04	1106.90	C2	C2
21	1300	273	794.89	17.81	1157.26	C2	C2
22	1280	269	774.49	3.27	1136.87	C2	C2
23	1380	276	874.12	97.35	1236.23	C2	C2
24	1150	253	643.84	133.81	1006.43	C2	C2
25	1420	284	914.90	137.97	1277.02	C2	C2
26	220	44	309.76	1086.23	53.25	C3	C3
27	250	50	279.24	1055.69	83.85	C3	C3
28	200	40	330.11	1106.68	32.86	C3	C3
29	680	163	165.43	612.02	528.32	C1	C1
30	720	173	206.66.	570.91	569.55	C1	C1

Using these updated centroids, the Euclidean distance from each of the 30 data points to all three centroids was recomputed. Each point was then reassigned to its nearest centroid. As shown in the Classification column, no data point changed its cluster assignment from Iteration 1, all 30 points retained the same classification.

Since no data point changed its cluster assignment between Iteration 1 and Iteration 2, the algorithm has reached convergence. This means the centroids have stabilized and further iterations would produce no change. The k-means algorithm terminates here with the following final clusters:

- Cluster 1 (C1) – Medium-weight, mid-priced books (points 7–15, 29, 30)
- Cluster 2 (C2) – Heavy, high-priced books (points 16–25)
- Cluster 3 (C3) – Lightweight, low-priced books (points 1–6, 26–28)

These three clusters confirm the optimal k = 3 determined by the Elbow Method, validating that the dataset naturally separates into three distinct groups based on weight and price.